

# Shruti A. Panicker

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Seeking full-time opportunities from May 2026 | Open to remote

## EDUCATION

**M.S. in Artificial Intelligence – ML Concentration**, *Northeastern University* *Boston, MA, USA* | Sep 2024 – May 2026  
**GPA:** 3.875/4 | **Coursework:** Foundations of AI, Data Processing for AI, Machine Learning, NLP, Human Computer Interaction

**B.Tech in Computer Science and Engineering**, *Amrita Vishwa Vidyapeetham* *Coimbatore, India* | Jul 2016 – Aug 2020  
**GPA:** 8.06/10 | **Coursework:** Data Structures, Computational Intelligence, Deep Learning, Data Science

## EXPERIENCE

**Graduate Teaching Assistant**, *Northeastern University* *Boston, MA, USA* | Sep 2024 – Apr 2026

- Led instruction for DS2500 across all 4 MS semesters (250+ students/semester), redesigning assignments and project structure subsequently adopted department wide.

**ML Engineer Consultant**, *Nexeralabs Pvt Ltd* *Trivandrum, India* | Jun 2024 – Aug 2024

- Reduced group discussion scoring time **from 1–2 hours to under 2 minutes** by building an agentic pipeline with automated transcription, speaker diarization, and LLM-based rubric scoring across up to 8 participants.
- Eliminated inconsistent mentor feedback by auto-generating per-speaker PDF reports scoring 10 evaluation criteria through a structured LLM-scoring architecture, deployed at scale.

**Data Scientist Lead**, *Mu Sigma Business Solutions* *Bangalore, India* | Jan 2024 – May 2024

- Directed a 10-person decision science team delivering product analytics for a Fortune 20 technology company's collaboration device portfolio, owning project lifecycle from problem definition to executive presentation.
- Built a statistical upsell propensity model in Python and SQL, applying TAM/SAM/SOM segmentation across large-scale datasets to identify high-conversion targets and **drive 5% product adoption growth** through focused campaigns.
- Automated 25+ product KPI dashboards in Python and SQL, eliminating all manual data pulls and enabling weekly data-driven decisions for product leadership.
- Translated analytical insights into product strategy and investment decisions by partnering with product and business leadership on C-suite presentations, directly influencing roadmap priorities.

**Decision Scientist**, *Mu Sigma Business Solutions* *Bangalore, India* | Sep 2020 – Dec 2023

- Built a graph-based entity resolution system in Python to deduplicate Device IDs across large-scale datasets, applying network association analysis to identify anomalous identity patterns and restore integrity across all downstream portfolio metrics.
- Designed a three-state customer lifecycle model (inactive/churned/reactivated) using classification and behavioral segmentation, enabling targeted re-engagement campaigns that **drove 18% engagement increase within 6 months**.
- Diagnosed product onboarding drop-off by joining behavioral usage signals with support data in Python and SQL, pinpointing friction points across onboarding stages and recommending targeted interventions that **drove a 30% increase in reactivation rates**.
- Led 8-month EUDB-compliant pipeline migration across 25+ dashboards, redesigning ETL transformation and aggregation layers to ensure EU data residency compliance with zero loss in metric accuracy.

**Data Analyst**, *Riveron Infotech Pvt Ltd* *Trivandrum, India* | Jan 2020 – Aug 2020

- Conducted cohort analysis tracking student progression across program levels, identifying drop-off points that drove targeted interventions and **improved program completion rates by 15%**.
- Built a churn prediction model using behavioral signals (login frequency, activity completion, mentor engagement) to flag students as at-risk, enabling proactive outreach that **improved retention by 30%**.

## PROJECTS

**Semantic Turn Detection for Voice AI** *Northeastern University*  
Built an intent-conditioned voice system that classifies utterances as transactional, informational, or reflective and dynamically adapts LLM ensemble weights per intent type. Developed annotation guidelines and a human evaluation methodology; system achieved Naturalness 6/7, Frustration 1.53/7, and Trust 5.7/7 on a 7-point Likert scale.

**Empathy Detection – WASSA 2024 Shared Task** *Northeastern University*  
Fine-tuned BERT for regression across 11K+ conversational data points to predict empathy, emotional polarity, and intensity. Designed intrinsic and extrinsic evaluation frameworks to validate model behavior, achieving 0.597 average Pearson correlation, which is a score that would have placed in the top 4 on the WASSA 2024 shared task leaderboard.

## SKILLS

**Languages & Query:** Python, SQL, R, C#, SCOPE, Power BI DAX, Git/GitHub | **ML & Modeling:** Pandas, NumPy, PyTorch, TensorFlow, Keras, scikit-learn, BERT/Transformers, Hugging Face, Statistical Modeling, Classification, Regression, Feature Engineering | **NLP & LLMs:** NLP, LLM Fine-tuning, LLM Evaluation, Agentic AI, Prompt Engineering, Responsible AI | **Data & Analytics:** A/B Testing, Causal Inference, Experimentation, User Behavior Analysis, Cohort Analysis, Funnel Analysis | **Data Engineering:** Apache Spark, PySpark, AWS, Azure, Hadoop | **Visualization & BI:** Power BI, Tableau | **Compliance:** EUDB & GDPR Compliance | **Certifications:** AWS Certified Cloud Practitioner, AWS Certified Machine Learning Engineer Associate (in progress)